

Question 6

IPO & PAC Chart

INPUT - PROCESS - OUTPUT

~~INPUT~~

- vehicle type
- Battery level
- Required charging
- Parking duration
- Membership
- Priority
- Station available

- compare battery levels
- compare stations
- calculate priority
- calculate rate after comparing season.
- calculate Parking duration.

- vehicle type
- current Battery
- Required Battery
- Priority message
- It can change or not
- Final cost

→ PSEUDOCODE

INPUT " E
INPUT " Bat
INPUT " Re
INPUT " Pa
INPUT " M
INPUT " P
INPUT "

OUTPUT
OUTPUT
OUTPUT

IF sta

END

IF



→ PSEUDOCODE

INPUT "Enter vehicle type", ~~vehicle~~ type

INPUT "Battery Level", Battery-lvl

INPUT "Required Level", Req-lvl

INPUT "Parking Duration", Park-durr

INPUT "Membership", membership

INPUT "Priority", Priority

INPUT "Station Available", Station-avail

OUTPUT "Vehicle Type", type

OUTPUT "Current Battery level", Battery-lvl, "%"

OUTPUT "Required Battery ~~level~~ level", Req-lvl, "%"

IF Station-avail == "N" THEN

IF type == "H" THEN

OUTPUT "Changing unavailable - parking available"

ELSE

OUTPUT "No changing slot"

ENDIF

CanChange ← FALSE

ENDIF

IF CanChange == TRUE THEN

Req-change = Req-lvl - Battery-lvl

IF Req-lvl ≤ Battery-lvl THEN

Date: _____

```

    OUTPUT "No charging required"
    cancharge ← FALSE
  ENDIF
ENDIF

IF cancharge == TRUE THEN
  IF battery-lvl ≤ 15 AND req-lvl ≥ 80 THEN
    Priority = "Emergency Charging Priority"
  ELSEIF disabled-Priority == "Y" OR (membership == "Y" AND
    Battery-lvl ≤ 30) THEN
    Priority = "Priority Charging"
  Else
    Priority = "Normal Charging"
  ENDIF
ENDIF

OUTPUT "Charging Priority", Priority

IF time ≥ 17 AND time < 22 THEN
  Peak = Peak "Peak"
  Unitrate = 50
  dist-Peak = 0.1
ELSE
  Peak = "Off-Peak"
  Unitrate = 35
  dist-Peak = 0.2
ENDIF
  
```

Date: _____

OUTPUT "Time Status", Peak

Charging Cost = Req-Charge * Unitrate

Chargingdist = 0

IF Priority = "Emergency Charging" THEN

~~chargingcost~~

Final cost ← charging cost

ELSE

IF membership == "Y" THEN

IF Peak = "Peak" THEN

chargingdist = Charging Cost * ~~1000~~ dist - Peak

Else

Charging dist = Charging cost * dist - offpeak

ENDIF

ENDIF

Final Cost ← charging cost - charging dist

ENDIF

IF ~~Compare~~ Park-dur ≤ 2 THEN

BasePcost = 200

ElseIF Park-dur ≤ 5 THEN

BasePcost = 400

Else

Base Pcost = 700

ENDIF



Date: _____

Park dist = 0

IF disabled priority == "Y" THEN

ParkingCost = 0

Parkingdist = basePost

ELSE

IF membership == "Y" THEN

Parkingdist = basePost * 0.2

ENDIF

ParkingCost = basePost - Parkingdist

ENDIF

dist = + Parkingdist

~~IF Park-dist > 8 THEN~~

IF Park-dist > 8 THEN

OUTPUT "Relocate your vehicle after changing"

Else

OUTPUT "standard Parking Duration"

ENDIF

FinalAmount = ~~basePost~~ Finalcost + Parking Cost

OUTPUT Final Cost , Parking Cost , dist , FinalAmount